Nederlandse norm

NEN-EN 1005-4+A1

(en)

Safety of machinery - Human physical performance - Part 4: Evaluation of working postures and movements in relation to machinery

Vervangt NEN-EN 1005-4:2005

ICS 13.110-30; 13.180

oktober 2008

Dit document mag slechts op een stand-alone PC worden geïnstalleerd. Gebruik op een netwerk is alleen toegestaan als een aanvullende licentieovereenkomst voor netwerkgebruik met NEN is afgesloten.
This document may only be used on a stand-alone PC. Use in a network is only permitted when a supplementary license agreement for use in a network with NEN has been concluded.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Als Nederlandse norm is aanvaard:
- EN 1005-4:2005+A1:2008,IDT

Normcommissie 302 020 "Ergonomie Platform"

Apart from exceptions provided by the law, nothing from this publication may be duplicated and/or published by means of photocopy, microfilm, storage in computer files or otherwise, which also applies to full or partial processing, without the written consent of the Netherlands Standardization Institute.

The Netherlands Standardization Institute shall, with the exclusion of any other beneficiary, collect payments owed by third parties for duplication and/or act in and out of law, where this authority is not transferred or falls by right to the Reproduction Rights Foundation.

Although the utmost care has been taken with this publication, errors and omissions cannot be entirely excluded. The Netherlands Standardization Institute and/or the members of the committees therefore accept no liability, not even for direct or indirect damage, occurring due to or in relation with the application of publications issued by the Netherlands Standardization Institute.

Auteursrecht voorbehouden. Behoudens uitzondering door de wet gesteld mag zonder schriftelijke toestemming van het Nederlands Normalisatie-instituut niets uit deze uitgave worden verveelvoudigd en/of openbaar gemaakt door middel van fotokopie, microfilm, opslag in computerbestanden of anderszins, hetgeen ook van toepassing is op gehele of gedeeltelijke bewerking.

Het Nederlands Normalisatie-instituut is met uitsluiting van ieder ander gerechtigd de door derden verschuldigde vergoedingen voor verveelvoudiging te innen en/of daartoe in en buiten rechte op te treden, voor zover deze bevoegdheid niet is overgedragen c.q. rechtens toekomt aan de Stichting Reprorecht.

Hoewel bij deze uitgave de uiterste zorg is nagestreefd, kunnen fouten en onvolledigheden niet geheel worden uitgesloten. Het Nederlands Normalisatie-instituut en/of de leden van de commissies aanvaarden derhalve geen enkele aansprakelijkheid, ook niet voor directe of indirecte schade, ontstaan door of verband houdend met toepassing van door het Nederlands Normalisatie-instituut gepubliceerde uitgaven.

©2008 Nederlands Normalisatie-instituut
Postbus 5059, 2600 GB Delft
Telefoon (015) 2 690 390, Fax (015) 2 690 190

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Nederlands voorwoord

Voor de in deze norm vermelde normatieve verwijzingen bestaan in Nederland de volgende equivalenten:

vermelde norm	Nederlandse norm	titel
EN 547-1	NEN-EN 547-1	Veiligheid van machines - Menselijke lichaamsafmetingen - Deel 1: Principes voor de bepaling van de vereiste afmetingen van toegangsopeningen in machines voor het gehele lichaam (en,nl)
EN 547-2	NEN-EN 547-2	Veiligheid van machines - Menselijke lichaamsafmetingen - Deel 2: Principes voor de bepaling van de vereiste afmetingen van toegangsopeningen (en,nl)
EN 547-3	NEN-EN 547-3	Veiligheid van machines - Menselijke lichaamsafmetingen - Deel 3: Antropometrische gegevens (en,nl)
EN 614-1	NEN-EN 614-1	Veiligheid van machines - Ergonomische ontwerpprincipes - Deel 1: Terminologie en algemene principes (en)
EN 894-1	NEN-EN 894-1	Veiligheid van machines - Ergonomische eisen voor het ontwerpen van informatie- en bedieningsmiddelen - Deel 1: Algemene beginselen voor de interactie tussen de mens en informatie- en bedieningsmiddelen (en,nl)
EN 894-2	NEN-EN 894-2	Veiligheid van machines - Ergonomische eisen voor het ontwerpen van informatie- en bedieningsmiddelen - Deel 2: Informatiemiddelen (en,nl)
EN 894-3	NEN-EN 894-3	Veiligheid van machines - Ergonomische eisen voor het ontwerpen van informatie- en bedieningsmiddelen - Deel 3: Bedieningsmiddelen (en)
EN 1005-1:2001	NEN-EN 1005-1:2001	Veiligheid van machines - Menselijke fysieke belasting - Deel 1: Termen en definities (en)
EN 1005-2	NEN-EN 1005-2	Veiligheid van machines - Menselijke fysieke belasting - Deel 2: Handmatig hanteren van machines en machine-onderdelen (en)
EN 1005-3	NEN-EN 1005-3	Veiligheid van machines - Menselijke fysieke belasting - Deel 3: Aanbevolen maximale krachten bij machinewerkzaamheden (en)
prEN 1005-52	-	-
EN 1050	-	-
EN ISO 12100-1:2003	NEN-EN-ISO 12100-1:2003	Veiligheid van machines - Basisbegrippen, algemene ontwerpbeginselen - Deel 1: Basisterminologie, methodologie (en,nl)
EN ISO 12100-2	NEN-EN-ISO 12100-2	Veiligheid van machines - Basisbegrippen, algemene ontwerpbeginselen - Deel 2: Technische beginselen (en)
EN ISO 14738	NEN-EN-ISO 14738	Veiligheid van machines - Antropometrische eisen voor het ontwerp van werkplekken bij machines (en)

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN 1005-4:2005+A1

October 2008

ICS 13.110; 13.180

Supersedes EN 1005-4:2005

English Version

Safety of machinery - Human physical performance - Part 4: Evaluation of working postures and movements in relation to machinery

Sécurité des machines - Performance physique humaine - Partie 4: Evaluation des postures et mouvements lors du travail en relation avec les machines

Sicherheit von Maschinen - Menschliche körperliche Leistung - Teil 4: Bewertung von Körperhaltungen und Bewegungen bei der Arbeit an Maschinen

This European Standard was approved by CEN on 17 February 2005 and includes Amendment 1 approved by CEN on 18 August 2008.

CEN members are bound to comply with the CEN/CENELEC Internal Regulations which stipulate the conditions for giving this European Standard the status of a national standard without any alteration. Up-to-date lists and bibliographical references concerning such national standards may be obtained on application to the CEN Management Centre or to any CEN member.

This European Standard exists in three official versions (English, French, German). A version in any other language made by translation under the responsibility of a CEN member into its own language and notified to the CEN Management Centre has the same status as the official versions.

CEN members are the national standards bodies of Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

EUROPEAN COMMITTEE FOR STANDARDIZATION
COMITÉ EUROPÉEN DE NORMALISATION
EUROPÄISCHES KOMITEE FÜR NORMUNG

Management Centre: rue de Stassart, 36 B-1050 Brussels

© 2008 CEN All rights of exploitation in any form and by any means reserved worldwide for CEN national Members.

Ref. No. EN 1005-4:2005+A1:2008: E

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Contents

Page

Foreword 3

Introduction 4

1 Scope 5

2 Normative references 5

3 Terms and definitions 6

4 Requirements 6
4.1 General 6
4.2 Guidance towards risk assessment 6
4.2.1 General 6
4.2.2 Establish the operator population 7
4.2.3 Perform a task analysis 7
4.2.4 Identify the ergonomic data required 7
4.2.5 Evaluate at the drawing-table/CAD-screen 8
4.2.6 Evaluate with operators 8
4.3 Risk assessment 8
4.3.1 General 8
4.3.2 Trunk 10
4.3.3 Upper arm 14
4.3.4 Head and neck 15
4.3.5 Other body parts 18

Annex ZA (informative) Relationship between this European Standard and the Essential Requirements of EU Directive 98/79/EC 20

Annex ZB (informative) A1 Relationship between this European Standard and the Essential Requirements of EU Directive 2006/42/EC A1 21

Bibliography 22

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Foreword

This document (EN 1005-4:2005+A1:2008) has been prepared by Technical Committee CEN/TC 122 “Ergonomics”, the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by April 2009, and conflicting national standards shall be withdrawn at the latest by December 2009.

This document includes Amendment 1, approved by CEN on 2008-08-18.

This document supersedes EN 1005-4:2005.

The start and finish of text introduced or altered by amendment is indicated in the text by tags ▌A1⟩ ⟨A1▌.

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

▌A1⟩ For relationship with EU Directive(s), see informative Annexes ZA and ZB, which are integral parts of this document. ⟨A1▌

EN 1005 consists of the following Parts, under the general title *Safety of machinery - Human physical performance*:

— *Part 1: Terms and definitions*;

— *Part 2: Manual handling of machinery and component parts of machinery;*

— *Part 3: Recommended force limits for machinery operation*;

— *Part 4: Evaluation of working postures and movements in relation to machinery*;

— *Part 5*[1] *: Risk assessment for repetitive handling at high frequency*.

This European Standard includes a Bibliography.

According to the CEN/CENELEC Internal Regulations, the national standards organizations of the following countries are bound to implement this European Standard: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Netherlands, Norway, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, Switzerland and United Kingdom.

1 This European Standard is under preparation by CEN/TC 122/WG 4 "Biomechanics".

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Introduction

About one third of all workers in the European Union are involved in painful or tiring postures for more than half of their working day, and close to 50 % of all workers are exposed to short repetitive tasks, which are mostly accompanied by painful and tiring movements [2]. Pain and fatigue may lead to musculoskeletal disorder, reduced productivity, and deteriorated posture and movement control. The latter can increase the risk of errors and may result in reduced quality and hazardous situations. Within the life cycle of a machine, from construction to dismantling, all machine-related actions require certain postures and movements. The role of the machinery designer should be to avoid painful and tiring postures and movements.

The requirements in this European Standard aim to reduce the health risks associated with machine-related actions and could also have a positive effect on the quality, efficiency and profitability of those actions.

The requirements in this European Standard are based on current ergonomic knowledge and expert opinions, and will be subject to changes in accordance with future research [1].

This document is a type B standard as stated in EN ISO 12100-1.

The provisions of this European Standard can be supplemented or modified by a type C standard.

For machines which are covered by the scope of a type C standard and which have been designed and built according to the provisions of that standard, the provisions of that type C standard take precedence over the provisions of this type B standard.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

1 Scope

This European Standard presents guidance when designing machinery or its component parts in assessing and affecting health risks due only to machine-related postures and movements, i.e. during assembly, installation, operation, adjustment, maintenance, cleaning, repair, transport, and dismantlement. This European Standard specifies requirements for postures and movements without any or with only minimal external force exertion. The requirements are intended to reduce the health risks for nearly all healthy adults.

This European Standard is not applicable to the machinery, which is manufactured before the date of publication of this European Standard by CEN.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 547-1, *Safety of machinery - Human body measurements - Part 1: Principles for determining the dimensions required for openings for whole body access into machinery.*

EN 547-2, *Safety of machinery - Human body measurements - Part 2: Principles for determining the dimensions required for access openings.*

EN 547-3, *Safety of machinery - Human body measurements - Part 3: Anthropometric data.*

EN 614-1, *Safety of machinery - Ergonomic design principles - Part 1: Terminology and general principles.*

EN 894-1, *Safety of machinery - Ergonomics requirements for the design of displays and control actuators - Part 1: General principles for human interactions with displays and control actuators.*

EN 894-2, *Safety of machinery - Ergonomics requirements for the design of displays and control actuators - Part 2: Displays.*

EN 894-3, *Safety of machinery - Ergonomics requirements for the design of displays and control actuators - Part 3: Control actuators.*

EN 1005-1:2001, *Safety of machinery - Human physical performance - Part 1: Terms and definitions.*

EN 1005-2, *Safety of machinery - Human physical performance - Part 2: Manual handling of machinery and component parts of machinery.*

EN 1005-3, *Safety of machinery - Human physical performance - Part 3: Recommended force limits for machinery operation.*

prEN 1005-5[2], *Safety of machinery - Human physical performance - Part 5: Risk assessment for repetitive handling at high frequency.*

EN 1050, *Safety of machinery – Principles for risk assessment.*

EN ISO 12100-1:2003, *Safety of machinery - Basic concepts, general principles for design - Part 1: Basic terminology, methodology (ISO 12100-1:2003).*

EN ISO 12100-2, *Safety of machinery - Basic concepts, general principles for design - Part 2: Technical principles (ISO 12100-2:2003).*

---

2 This European Standard is under preparation by CEN/TC 122/WG 4 "Biomechanics".

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

EN ISO 14738, *Safety of machinery - Anthropometric requirements for the design of workstations at machinery. (ISO 14738:2002).*

ISO 11226:2000, *Ergonomics - Evaluation of static working postures.*

3 Terms and definitions

For the purposes of this European Standard, the terms and definitions given in EN 1005-1:2001 and EN ISO 12100-1:2003 apply.

4 Requirements

4.1 General

It is recommended that those using this European Standard should first refer to EN 1050, EN ISO 12100-1, EN ISO 12100-2, and EN 614-1.

The requirements conveyed by this European Standard formally apply to all machine-related actions. Assessing each individual action may be impracticable, therefore those actions which are frequently undertaken or are infrequent but may give rise to risk of pain, fatigue or disorder shall be part of the risk assessment.

NOTE It is emphasised that the requirements in this European Standard should not be used to regulate work organisation.

Above all, machinery design shall allow for **variation** between and while sitting, standing, and walking. Awkward body postures and movements shall be avoided (e.g. kneeling, crouching). Good design shall encourage low frequency movements and avoid painful and tiring postures and high frequency movements (see 4.3).

This European Standard adopts a stepwise risk assessment approach for assessing postures and movements as part of the machinery design process (see Figure 1). The approach detailed in this European Standard makes a distinction between:

— Evaluation **without** operators: When there is no full-size model/prototype of the machinery or its parts currently available (see 4.2.5);

— Evaluation **with** operators: When a full-size model/prototype of the machinery or its parts is available (see 4.2.6).

Clause 4.2 provides guidance during the various design stages. Clause 4.3 provides the risk assessment procedure for determining the acceptability of postures and movements.

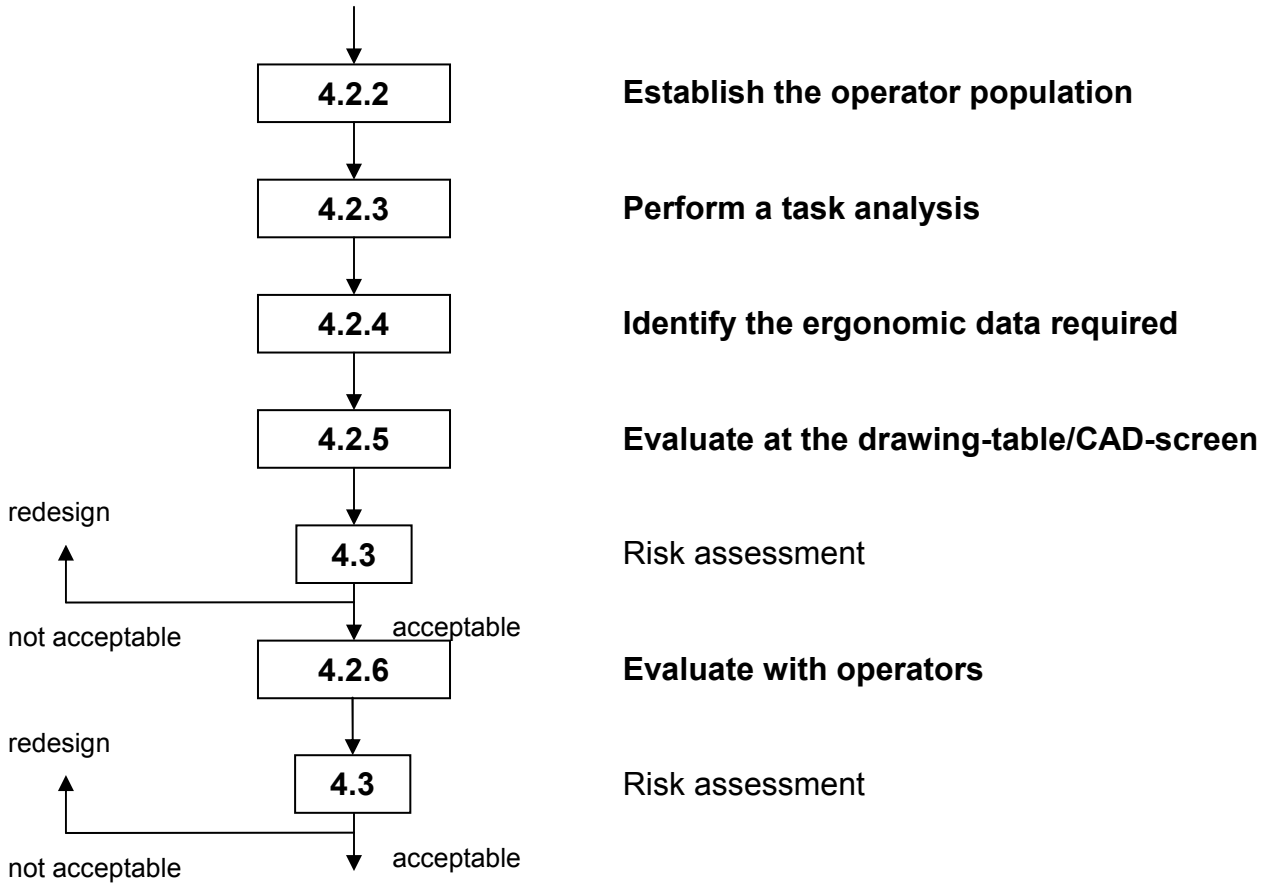
It is emphasised that there may be information available or obtainable on the use and consequences of particular designs and tasks performed. Provision shall be made for the collection and use of this data. This data shall be analysed to determine whether current designs and practices are adequate or whether a redesign of the machine is needed. The use of inexperienced subjects to perform established tasks may sometimes identify problems that experienced operators have learned to avoid.

4.2 Guidance towards risk assessment

4.2.1 General

Five main stages of the design process based on ergonomic tasks, as outlined in EN 614-1, are discussed in more detail (see 4.2.2 to 4.2.6 and Figure 1 (texts printed in bold)).

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44



NOTE The numbers in the boxes and the texts to the right refer to sections in this European Standard.

Figure 1 — Flow chart illustrating the risk assessment approach

4.2.2 Establish the operator population

A design shall accommodate the full range of intended operators. When considering the postures and movements of operators, it is important to determine the range of body dimensions of the operator population, the general principles of which are described in EN 614-1. The ergonomic tasks described in 4.2.4 to 4.2.6 rely on the selected body dimensions.

4.2.3 Perform a task analysis

Each task the operator is required to perform shall be identified and broken down into its individual components to produce a series of sequential events. All visual, control (hand/foot), stability, and force demands shall be determined for each of these events. The ergonomic tasks described in 4.2.4 to 4.2.6 are dependent upon this information. Furthermore, possible movement frequencies and work durations required with the machine shall be considered. The risk assessment described in 4.3.2 to 4.3.5 is dependent upon this information.

4.2.4 Identify the ergonomic data required

When considering the postures and movements of operators, EN ISO 14738 provides the ergonomic data required in any design process, i.e., with respect to:

— Selecting the main working posture
(sitting, standing, standing with a buttock rest);

— Machinery dimensions
(space/area demands, work surface height/slope).

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

This data alone will only establish the basic conditions for favourable postures and movements. Postures and movements may also be affected by the task demands listed below and these shall be considered early in the design process:

— Visual demands:
  Posture is adversely affected by an obstruction falling within the line of sight, low illumination levels, small-sized visual targets, and a small angle between the line of sight (gaze direction) and the surface of the visual target. Prolonged observation of a visual target to the side of the operator will lead to sustained head rotation and possible discomfort. In the vertical plane, the most favourable upward/downward direction of the line of sight depends upon the posture of the trunk, i.e., when leaning forwards the most favourable line of sight is in a more downwards, whereas when leaning backwards (i.e. against a backrest), it is in a more upwards direction;

— Control demands (operating a control device by hand/foot):
  Posture or movement may be adversely affected by the type and orientation of a hand grip/contact or pedal. For example, inappropriate orientation of the hand grip axis may create an unfavourable elevated upper arm;

— Stability demands:
  Posture may be affected by the need to support the body, or body parts, at the machinery. For example, when undertaking precision work, a low surface used to support the elbows can cause excessive forward bending of the trunk.

When there is a need to exert a force, reference shall be made to EN 1005-2, EN 1005-3, and prEN 1005-5. EN 894-1, EN 894-2, and EN 894-3 provide ergonomics requirements for the design of displays and control actuators.

4.2.5 Evaluate at the drawing-table/CAD-screen

Early in the design process a comparison shall be made between the body dimensions of the intended operator population (see 4.2.2) and the machinery dimensions. A number of methods may be used: standards (e.g. EN 547-1, EN 547-2, EN 547-3, EN ISO 14738), body templates or computer manikins.

It is recommended that a template or manikin be positioned in natural postures to simulate tasks, and that the risk assessment described in 4.3 of this European Standard be performed. If the design turns out to be ‘not acceptable', the user of this European Standard should go back to an earlier design stage and modify the design. This procedure should be repeated as many times as necessary, before going to 4.2.6.

4.2.6 Evaluate with operators

It is required to simulate tasks at least once by actual operators (selected test subjects representing the intended operator population) with a full-size model/prototype of the machinery or the machinery itself (see EN 614-1, ergonomic task 'evaluate with operators') and perform the risk assessment described in 4.3 of this European Standard. If the design turns out to be not acceptable, the user of this European Standard shall go back to an earlier design stage and modify the design.

4.3 Risk assessment

4.3.1 General

This European Standard uses a number of zones to evaluate postures and movements (see 4.3.2 to 4.3.5); three outcomes are possible:

— Acceptable:
  The health risk is considered low or negligible for nearly all healthy adults. No action is needed;

— Conditionally acceptable:
  There exists an increased health risk for the whole or part of the operator population. The risk shall be analysed together with contributing risk factors, followed as soon as possible by a reduction of the risks (i.e. redesign) or if that is not possible, other suitable measures shall be taken, for example the provision of operator guidelines to ensure that the use of the machine is acceptable;

— Not acceptable:

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

The health risk cannot be accepted for any part of the operator population. Redesign to improve the working posture is mandatory.

For the Tables 1 to 5 more than one posture and movement condition is considered acceptable. However, where acceptable appears in upper case, this is the recommended condition. ISO 11226 contains a detailed description of the procedure for determination of postures and movements (see Annex A).

NOTE 1 Current knowledge allows partially quantitative requirements, i.e., there may be non-acceptable combinations of otherwise acceptable postures or movements. For example, 10° bending forward of the trunk is acceptable according to the requirements (see 4.3.2.2). Not clearly visible twisting of the trunk (e.g. 5°) is acceptable according to the requirements (see 4.3.2.3). However, for a combination both positions quantitative requirements (i.e., is this acceptable or not) cannot yet be provided.

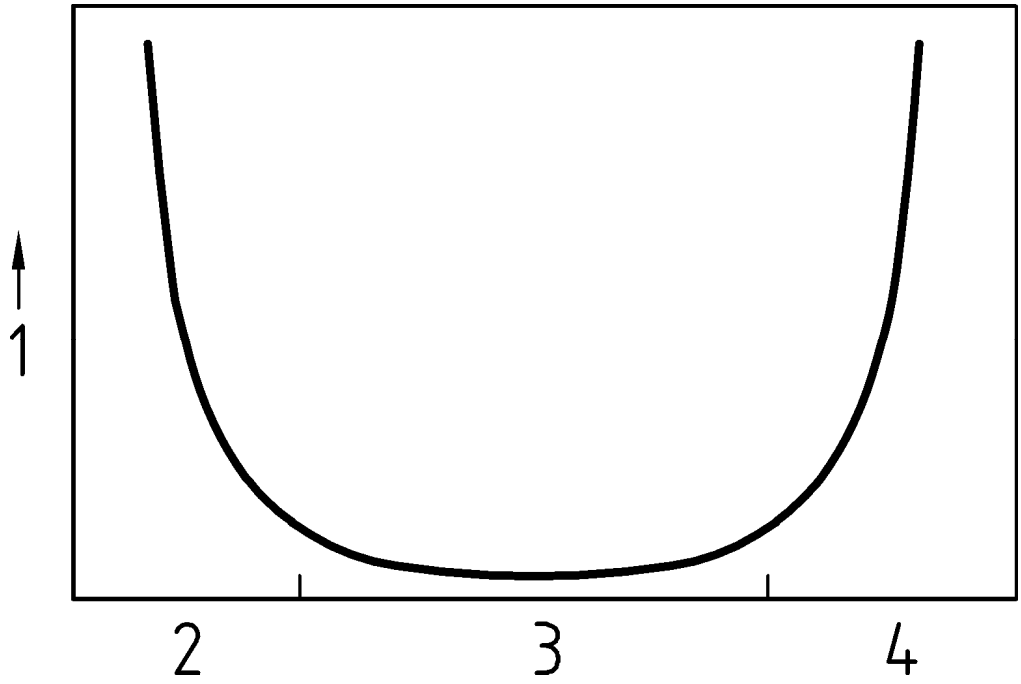
In 4.3.2 to 4.3.4 the above evaluation procedure constitutes only a first step if the postures or movements observed are assigned the outcome 'conditionally acceptable'. For this reason, Step 2 of the evaluation procedure is introduced. For particular postures, acceptability depends upon the nature and duration of the posture and period of recovery. Reference to ISO 11226 may be used as a guide. Furthermore, for other postures and movements acceptability may depend upon the frequency of the movement or the presence or absence of body support. It is emphasised that support of the arm or the trunk by machinery (directly or indirectly by the arm) may lead to different evaluation results for the upper arm, trunk and head. In this case, refer also to ISO 11226. Finally, it should be recognised that personal protective equipment (for instance, think about extra weight, increased body dimensions, and restricted range-of-motion) or tools (e.g. extra heavy) may affect the evaluation result.

Working periods of long duration and high movement frequencies are known to increase health risks due to machine-related working postures and movements. Current knowledge only allows partial quantitative evaluation of these risk factors (see Step 2 of the evaluation procedure in 4.3.2 to 4.3.4). Therefore, particularly if the machine may be used under the conditions mentioned above, it is strongly recommended that maximum improvement of the working posture be achieved, even if the working posture or movement is already assigned the outcome 'acceptable'. It is required that a text on this issue be incorporated in the user manual of the machine, e.g.: ‘If this machine or similar is used by the same operator for a considerable part of the working day as part of the daily work routine and requires static posture without adequate recovery time or body support being provided, or requires high frequency movements, it may cause pain, fatigue, musculoskeletal disorder, and hazardous situations. It is recommended that a health and safety professional and a production engineer work closely together to assess the status of the operator and the machine.’

The risk assessment procedure detailed below is based on the U-shaped model which proposes that health risks increase when the task approaches either end of the curve, i.e., if there is little or no movement (denoted static posture), or if movement frequencies are high. See Figure 2. It is emphasised that the term ‘movement frequency’ refers to the number of particular movements of a body segment per minute, and not to the number of work cycles per minute. That is, one work cycle may contain more than one particular movement of a body segment.

NOTE 2 Due to the fact that musculoskeletal fatigue is a main factor limiting the performance of work covered by this European Standard, different evaluation results for particular movement frequencies may come out as compared to EN 1005-2 (concerning manual handling), which is based on an energy expenditure criterion, amongst others.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44



Key
1 Health risk
2 Static posture
3 Movement low frequency
4 Movement high frequency

Figure 2 — A model of the health risks associated with postures and movements

4.3.2 Trunk

4.3.2.1 General

Trunk posture shall be evaluated by considering trunk bending forward/backward (see 4.3.2.2), trunk bending sideways (see 4.3.2.3), and trunk twisting (see also 4.3.2.3). The procedure applies to sitting and standing.

4.3.2.2 Trunk bending forward/backward

In Step 1 below trunk bending forward/backward shall be determined and classified in one of the zones of Figure 3 (Clause A.2 of ISO 11226:2000 contains a detailed description of the procedure for determination of trunk posture and movement). For static posture, low movement frequencies, and high movement frequencies, whichever applies, Table 1 provides the evaluation outcomes for each zone. Should the outcome be 'conditionally acceptable', Step 2 (a, b, or c) below provides the conditions for assigning the outcome ‘acceptable’ or the outcome ‘not acceptable’.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Step 1

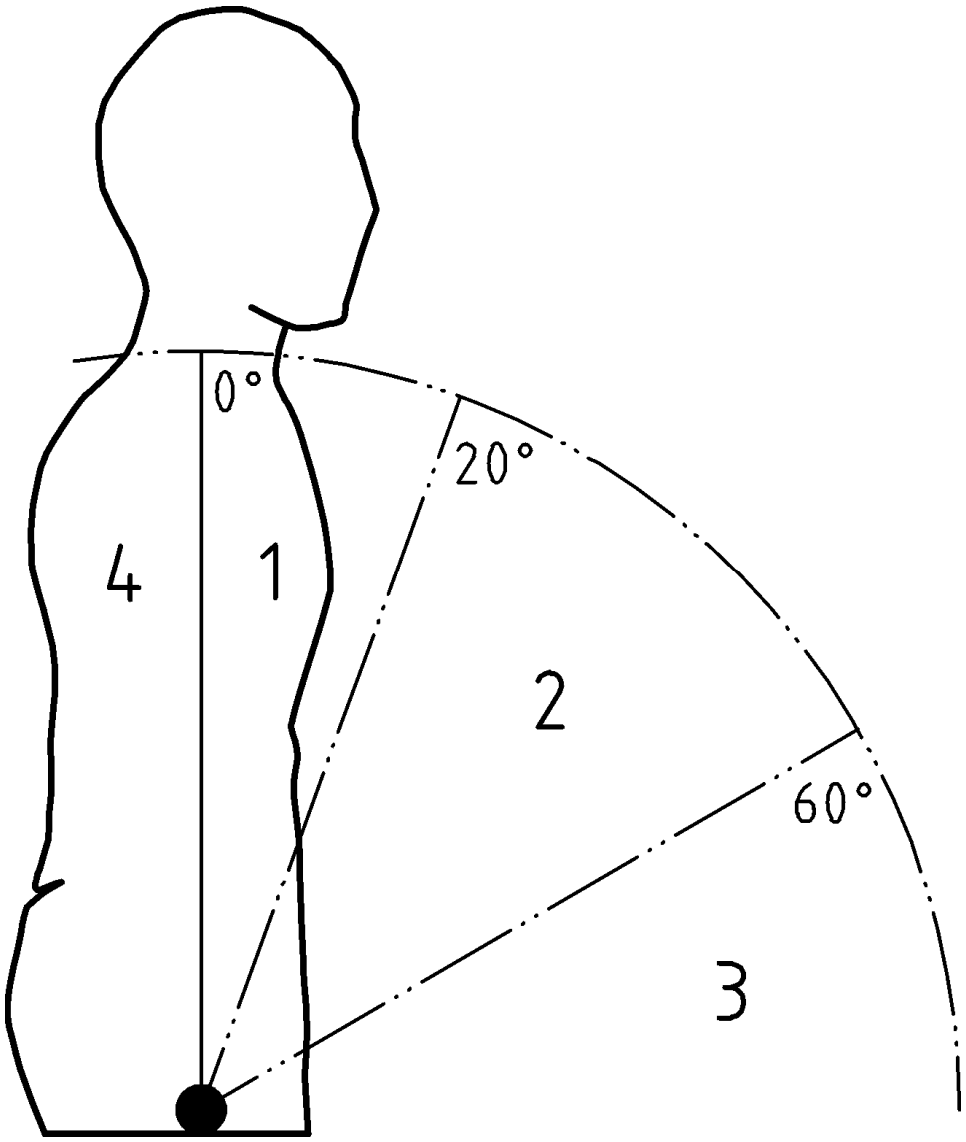


Figure 3 — Zones of trunk bending forward/backward

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Table 1 — Evaluation of trunk bending forward/backward

Zone	Static posture	Movement	
		Low frequency (< 2/min)	High frequency (≥ 2/min)
1[a]	Acceptable	ACCEPTABLE	Acceptable
2	Conditionally acceptable (Step 2a)	Acceptable	Not acceptable
3	Not acceptable	Conditionally acceptable (Step 2c))	Not acceptable
4	Conditionally acceptable (Step 2b)	Conditionally acceptable (Step 2c))	Not acceptable

[a] It is recommended that working postures with an upright trunk be achieved, in particular if the machine may be used for long durations by the same person, requiring static posture without adequate recovery time or body support being provided, or high frequency movements.

Step 2

a) Acceptable if there is full trunk support; if there is no full trunk support, acceptability depends upon duration of the posture and period of recovery. Full support during trunk bending forward is not acceptable, unless it is demonstrated that health risks are low or negligible for nearly all healthy adults considering the duration that the machine may be used.

b) Acceptable if there is full trunk support.

c) Not acceptable if the machine may be used for long durations by the same person. Exception: acceptable for low frequency movements into zone 4 if there is full trunk support. Full support during trunk bending forward is not acceptable, unless it is demonstrated that health risks are low or negligible for nearly all healthy adults considering the duration that the machine may be used.

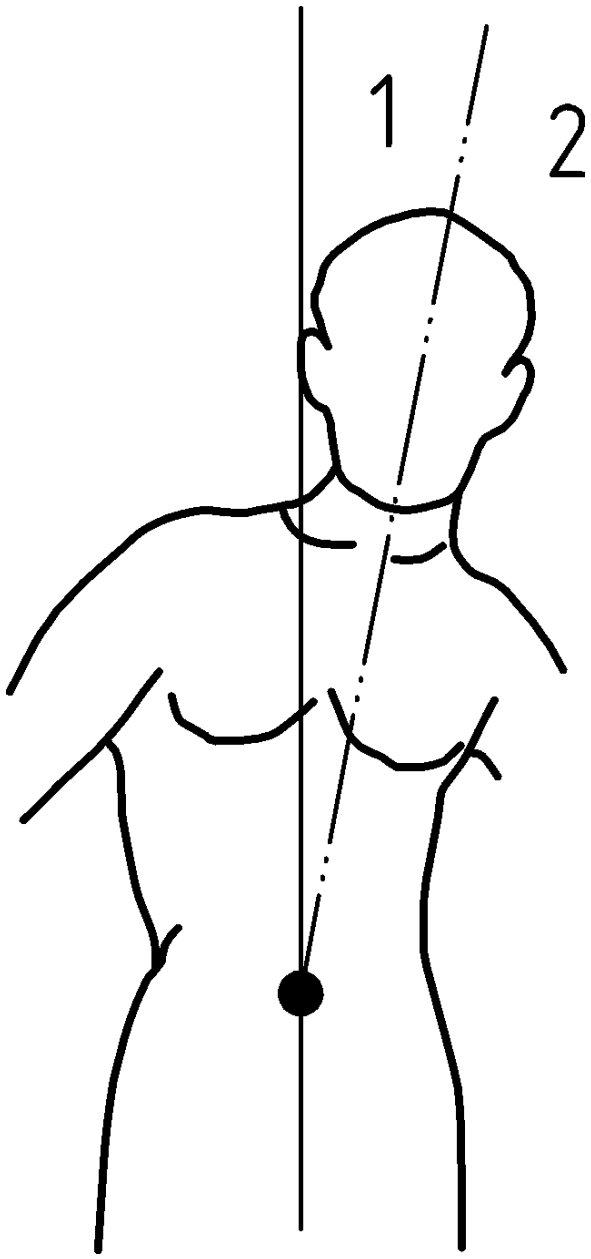
NOTE Full support of the trunk weight during trunk bending backward may for instance be provided by a high chair backrest. Full support during trunk bending forward may for instance be provided by a trunk harness or by leaning against a stable object, i.e., directly with the trunk, or indirectly by the arm(s) acting as a strut. Full trunk support in the examples mentioned may for instance hamper breathing, lead to undue local pressure, or induce shoulder and arm muscle fatigue.

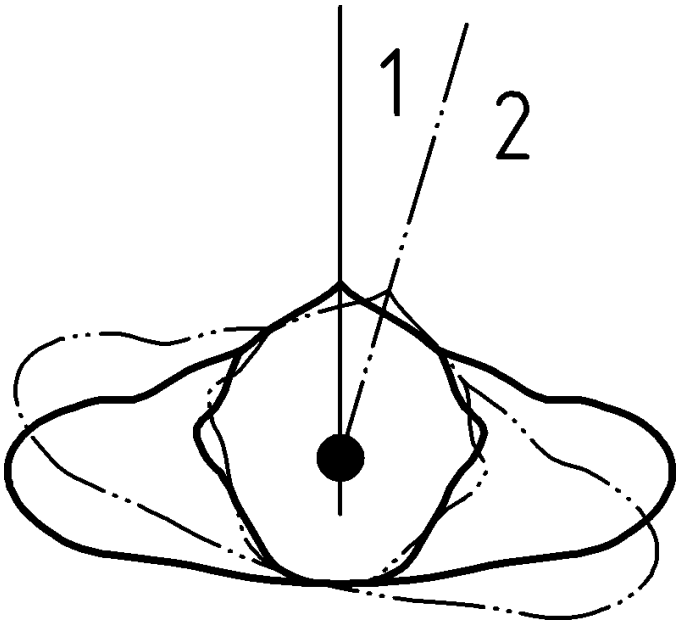
4.3.2.3 **Trunk bending sideways or twisting** (i.e. upper part of trunk with respect to lower part)

In Step 1 below trunk bending sideways (to the left or to the right), as well as trunk twisting (to the left or to the right) shall be determined and each classified into one of the zones of Figure 4 and Figure 5, respectively. For static posture, low movement frequencies and high movement frequencies (whichever applies), Table 2 provides the evaluation outcomes for each zone. In the case that the outcome is 'conditionally acceptable', Step 2a) below provides the conditions for assigning the outcome ‘acceptable’ or the outcome ‘not acceptable’.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Step 1





Key (Figures 4 and 5)

1 Trunk bending sideways or twisting not clearly visible (approximately 10° or less)

2 Trunk bending sideways or twisting clearly visible (approximately 10° or more)

The solid 'centre' lines represent an upright non-twisted trunk.

Figure 4 — Zones of trunk bending sideways

Figure 5 — Zones of trunk twisting

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Table 2 — Evaluation of trunk bending sideways or twisting

Zone	Static posture	Movement	
		Low frequency (< 2/min)	High frequency (≥ 2/min)
1	Acceptable	ACCEPTABLE	Acceptable
2	Not acceptable	Conditionally acceptable (Step 2a))	Not acceptable

Step 2

a) Not acceptable if the machine may be used for long durations by the same person.

4.3.3 Upper arm

In Step 1 below upper arm posture shall be determined and classified into one of the zones of Figure 6. (Clause A.3 of ISO 11226:2000 contains a detailed description of the procedure for determination of upper arm posture and movement). For static posture, low movement frequencies and high movement frequencies (whichever applies), Table 3 provides the evaluation outcomes for each zone. Should the outcome be 'conditionally acceptable', Step 2 (a), b), or c)) below provides the conditions for assigning the outcome ‘acceptable’ or the outcome ‘not acceptable’. The procedure applies to sitting and standing. For other upper extremity postures and movements see 4.3.5.

Step 1

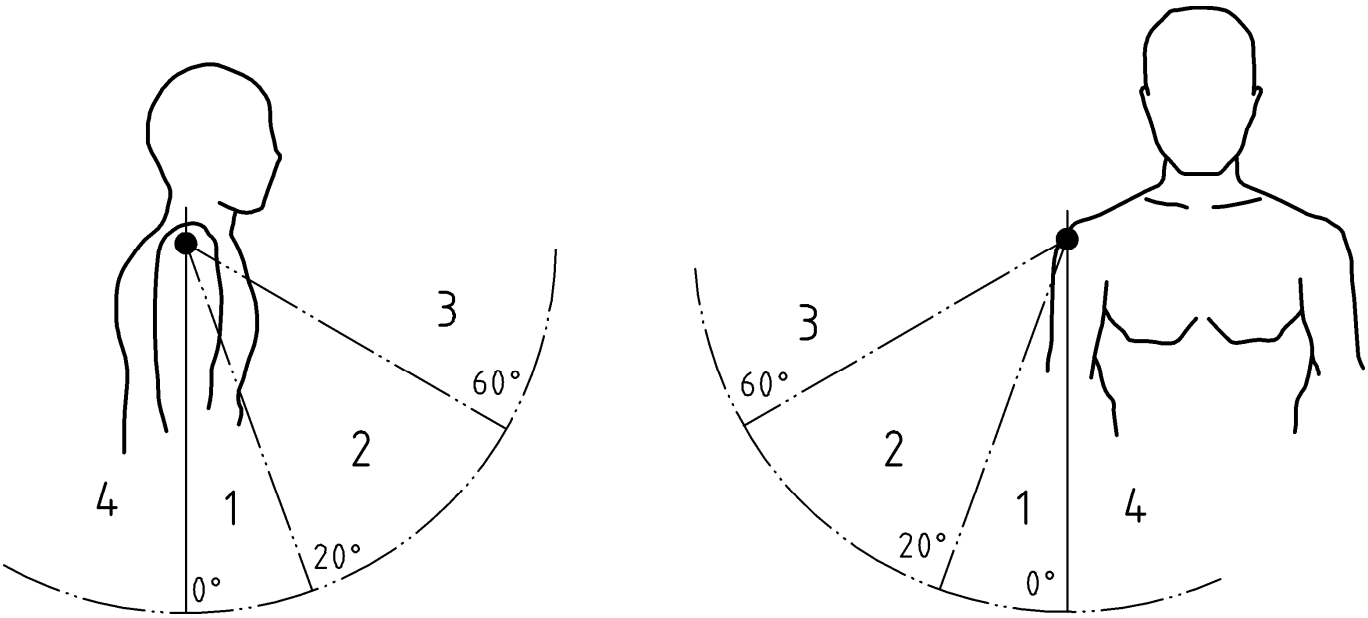


Figure 6 — Zones of upper arm posture

Zone 4 refers to the position of the upper arm with respect to the trunk, i.e., elbow behind the trunk when viewed from the side of the trunk (left part of Figure 6), or elbow not visible when viewed from behind the trunk (right part of Figure 6). Zones 1, 2, and 3 refer to the angle between the upper arm and the vertical, that is the ‘real’ angle, i.e., not dependent upon a viewing direction as for zone 4.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Table 3 — Evaluation of upper arm posture

Zone	Static posture	Movement	
		Low frequency (< 2/min)	High frequency (≥ 2/min)
1[a]	Acceptable	ACCEPTABLE	Acceptable
2	Conditionally acceptable (Step 2a))	Acceptable	Conditionally acceptable (Step 2c))
3	Not acceptable	Conditionally acceptable (Step 2b)	Not acceptable
4	Not acceptable	Conditionally acceptable (Step 2b)	Not acceptable

[a] It is recommended that working postures with the upper arms hanging down be achieved, in particular if the machine may be used for long durations by the same person, requiring static posture without adequate recovery time or body support being provided, or high frequency movements.

Step 2

a) Acceptable if there is full arm support; if there is no full arm support, acceptability depends upon the duration of the posture and period of recovery;

b) Not acceptable if the machine may be used for long durations by the same person;

c) Not acceptable if frequency ≥ 10/min or if the machine may be used for long durations by the same person.

NOTE Full support of the upper arm weight may for instance be achieved by placing the elbow/forearm on the machine. However, it should be pointed out that support of any kind may restrict free movement and cause localised pressure points.

4.3.4 Head and neck

4.3.4.1 General

Head and neck posture shall be evaluated by considering the upward/downward line of sight (gaze direction) (see 4.3.4.2), neck bending sideways (see 4.3.4.3), and neck twisting (see also 4.3.4.3). The procedure applies to sitting and standing.

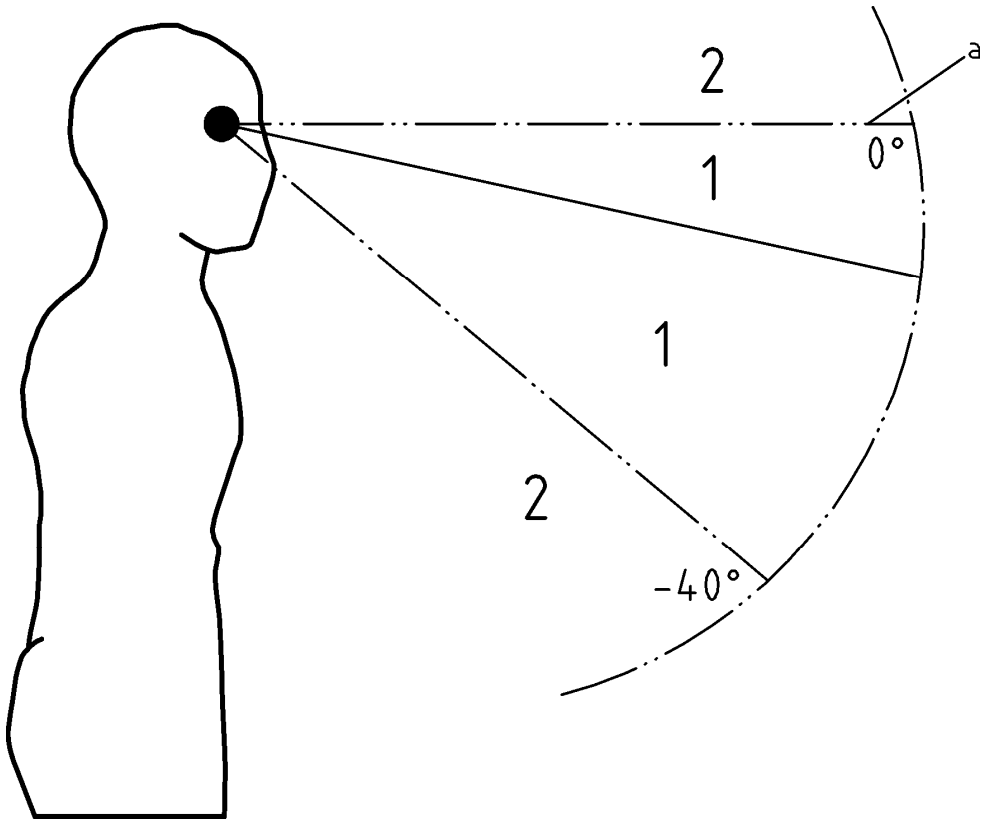
4.3.4.2 Upward/downward line of sight (gaze direction)

It is recommended that head and neck bending forward/backward be evaluated by the upward/downward line of sight (gaze direction), i.e. the centre of the field of vision (around the line of sight the eyes are able to view an area of between roughly 10° upwards and 10° downwards). The evaluation procedure shown below is a simplified version of the procedure described in ISO 11226:2000 (in the main text and in A.2). The procedure shown below is valid for an upright trunk posture (for other trunk postures other requirements apply).

In Step 1 below the upward/downward line of sight (gaze direction) shall be determined and classified into one of the zones of Figure 7. For static posture, low movement frequencies and high movement frequencies (whichever applies), Table 4 provides the evaluation outcomes for each zone. Should the outcome be 'conditionally acceptable', Step 2a below provides the conditions for assigning the outcome ‘acceptable’ or the outcome ‘not acceptable’.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Step 1



Key
a Horizontal

Figure 7 — Zones of upward/downward line of sight (gaze direction)

Table 4 — Evaluation of upward/downward line of sight (gaze direction)

Zone	Static posture	Movement	
		Low frequency (< 2/min)	High frequency (≥ 2/min)
1[a]	Acceptable	ACCEPTABLE	Acceptable
2	Not acceptable	Conditionally acceptable (Step 2a)	Not acceptable

[a] In the case of an upright trunk it is recommended that lines of sight (gaze directions) somewhat below the horizontal be achieved, in particular if the machine may be used for long durations by the same person, requiring static posture without adequate recovery time being provided, or high frequency movements.

Step 2

a) Not acceptable if the machine may be used for long durations by the same person.

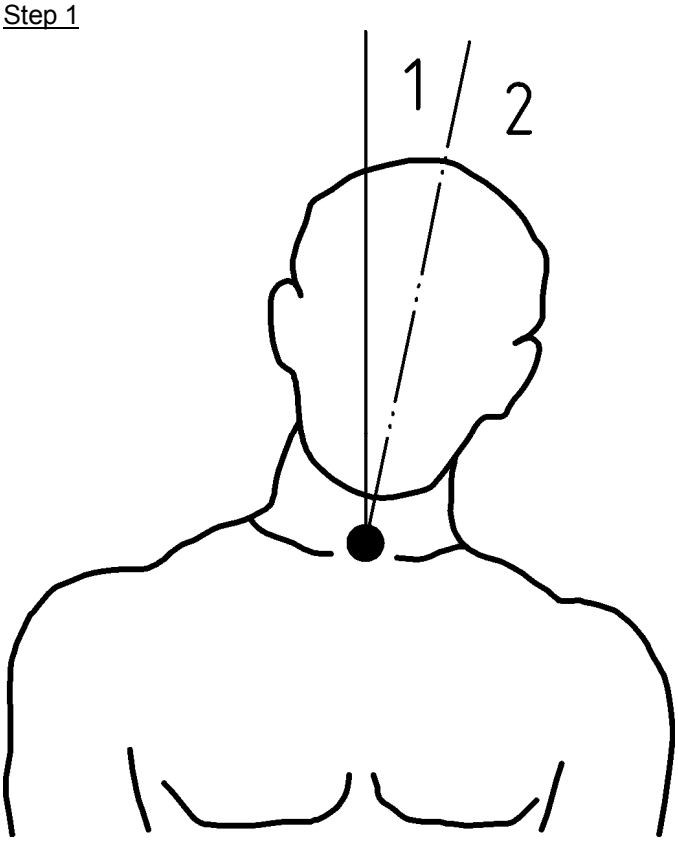
4.3.4.3 Neck bending sideways or twisting (i.e. head with respect to upper part of trunk)

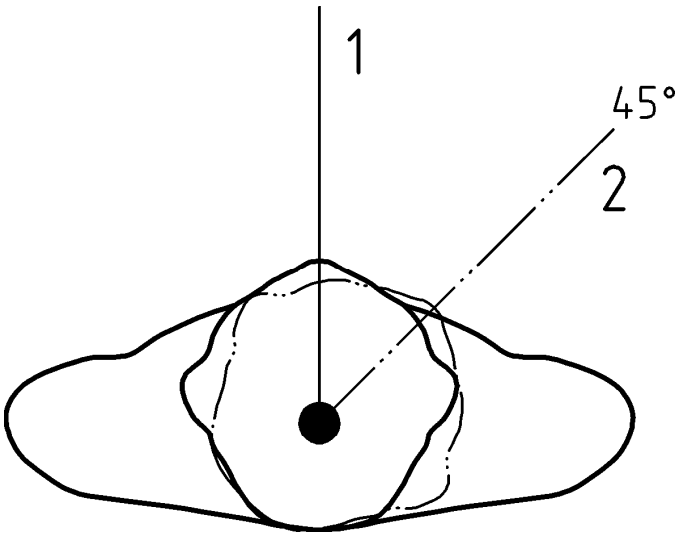
In Step 1 below neck bending sideways (to the left or to the right), as well as neck twisting (to the left or to the right) shall be determined and each classified into one of the zones of Figure 8 and Figure 9, respectively. For static posture, low movement frequencies and high movement frequencies (whichever applies), Table 5 provides the

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

evaluation outcomes for each zone. Should the outcome be 'conditionally acceptable', Step 2a below provides the conditions for assigning the outcome ‘acceptable’ or the outcome ‘not acceptable’.

Step 1





Key (Figure 8)

1 Not clearly visible (approximately 10° or less)

2 Clearly visible (approximately 10° or more)

The solid 'centre' line represents the neck in a non-bending sideways posture or movement.

The solid 'centre' line represents a non-twisted neck.

NOTE The solid and broken straight lines go trough the nose. The twisting of this line between 45° to the right may be used as a guidance for keeping neck twisting in zone 1.

Figure 8 — Zones of neck bending sideways

Figure 9 — Zones of neck twisting

Table 5 — Evaluation of neck bending sideways or twisting

Zone	Static posture	Movement	
		Low frequency (< 2/min)	High frequency (≥ 2/min)
1	Acceptable	ACCEPTABLE	Acceptable
2	Not acceptable	Conditionally acceptable (Step 2a)	Not acceptable

Step 2

a) Not acceptable if the machine may be used for long durations by the same person.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

4.3.5 Other body parts

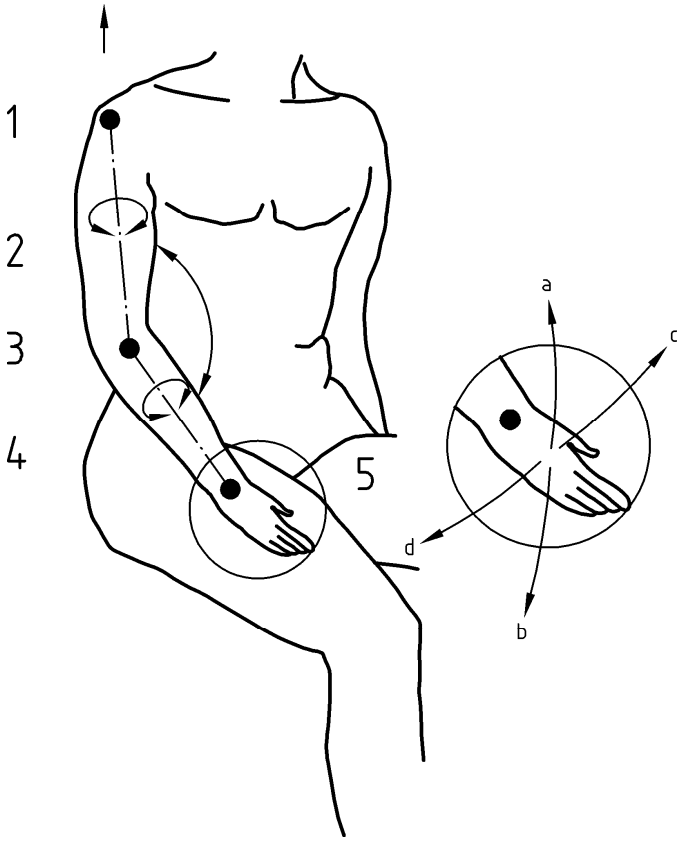
For static posture, low frequency movements and high frequency movements (whichever applies) of all other body parts (i.e., not evaluated in 4.3.2 to 4.3.4), Table 6 provides the evaluation outcomes. The procedure applies to sitting and standing, unless otherwise stated.

Table 6 — Evaluation of all other body parts

Static posture, i.e., awkward postures, such as convex low back (while sitting), extended knee and/or raised knee without the trunk leaning backward (while sitting), bent knee (while standing), raised shoulder, no even distribution of body weight over both feet (while standing) and joint positions near the limits of their range-of-motion [a] (for example, see Figures 10 and 11)	**Movement,** such as knee bending (while standing), shoulder raising, and joints nearing the limits of their range-of-motion [a] (for example, see Figures 10 and 11)	
	Low frequency (< 2/min)	High frequency (≥ 2/min)
Not acceptable	Acceptable	Not acceptable

[a] For joints, it is recommended that low frequency movements away from the limits of their range-of-motion be achieved. See A.4 in ISO 11226:2000 for a procedure for determining joint positions near the limit of the range-of-motion.

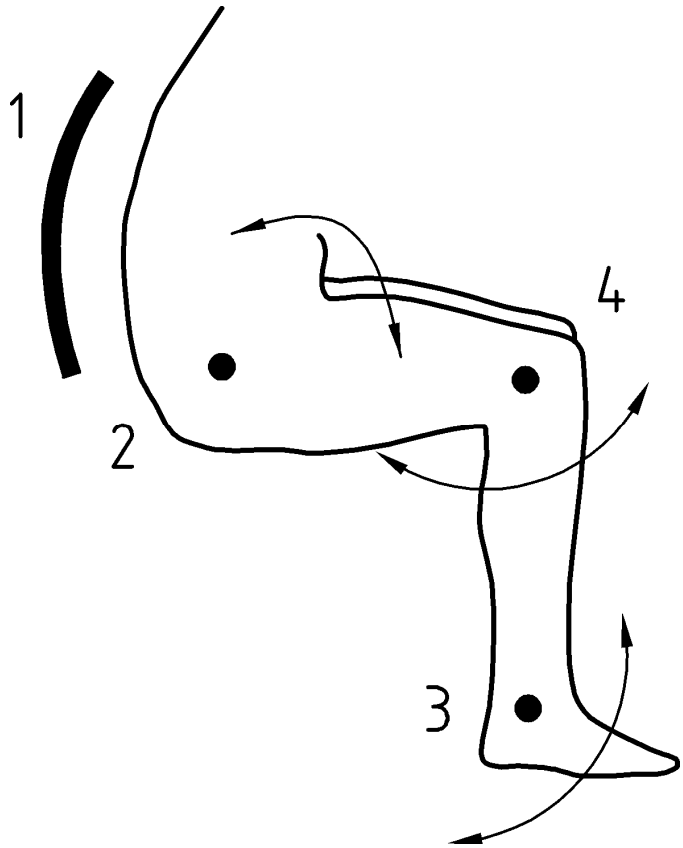
Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44



Key

1 Shoulder
2 Upper arm
3 Elbow
4 Forearm
5 Wrist
a Back of the hand moved towards the forearm
b Palm of the hand moved towards the forearm
c Thumb moved towards the forearm
d Little finger moved towards the forearm

Figure 10 — Upper extremities



Key

1 Low back
2 Hip
3 Ankle
4 Knee

Figure 11 — Low back and lower extremities

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Annex ZA
(informative)

Relationship between this European Standard and the Essential Requirements of EU Directive 98/79/EC

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide a means of conforming to Essential Requirements of the New Approach Directive 98/37/EC (Machinery), as amended by Directive 98/79/EC.

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the clauses of this standard given in Table ZA confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding Essential Requirements of that Directive and associated EFTA regulations.

Table ZA – Correspondence between this European Standard and Directive 98/37/EC

Clause(s)/sub-clause(s) of this European Standard	Essential Requirements (ERs) of Directive 98/37/EC	Qualifying remarks/Notes
Clause 4	1.1.2 d) 1.1.5	

WARNING — Other requirements and other EU Directives may be applicable to the product(s) falling within the scope of this standard.

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Annex ZB
(informative)

A1 Relationship between this European Standard and the Essential Requirements of EU Directive 2006/42/EC

This European Standard has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association to provide a means of conforming to Essential Requirements of the New Approach Directive 2006/42/EC on machinery.

Once this standard is cited in the Official Journal of the European Communities under that Directive and has been implemented as a national standard in at least one Member State, compliance with the normative clauses of this standard given in Table ZB.1 confers, within the limits of the scope of this standard, a presumption of conformity with the relevant Essential Requirements of that Directive and associated EFTA regulations.

Table ZB.1 — Correspondence between this European Standard and Directive 2006/42/EC

Clause(s)/sub-clause(s) of this EN	Essential Requirements (ERs) of Directive 2006/42/EC	Qualifying remarks/Notes
All clauses	Annex I: 1.1.5, 1.1.6	-

WARNING — Other requirements and other EU Directives may be applicable to the product(s) falling within the scope of this standard. A1

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44

Bibliography

[1] Delleman, N.J., Haslegrave, C.M., Chaffin, D.B.: Working Postures and Movements – Tools for Evaluation and Engineering. Taylor & Francis, London / CRC Press, Boca Raton, FL, USA, 2004.

[2] European Foundation for the Improvement of Living and Working Conditions 2001, Third European Survey on Working Conditions 2000 (Luxembourg: Office for Official Publications of the European Communities).

Dit document is door NEN onder licentie verstrekt aan: / This document has been supplied under license by NEN to:
politie.nl thijs.ter.avest@politie.nl 2025-12-16 19:28:44